

Chit 14 — MongoDB: Institute Database / Students Collection

1 & 2. Create database and collection

```
use Institute
// Collection is created automatically on first insert
```

3. Insert 10 documents

```
db.Students.insertMany([
  { RollNo:1, StudentName:'Amit', Age:20, Branch:'Computer',
    Address:{ City:'Pune', State:'Maharashtra' },
    Hobbies:['Cricket','Reading'] },
  { RollNo:2, StudentName:'Ankita', Age:18, Branch:'Electronics',
    Address:{ City:'Mumbai', State:'Maharashtra' },
    Hobbies:['Dancing','Painting'] },
  { RollNo:3, StudentName:'Ravi', Age:22, Branch:'Computer',
    Address:{ City:'Nashik', State:'Maharashtra' },
    Hobbies:['Gaming','Coding'] },
  { RollNo:4, StudentName:'Priya', Age:19, Branch:'Mechanical',
    Address:{ City:'Pune', State:'Maharashtra' },
    Hobbies:['Cooking','Singing'] },
  { RollNo:5, StudentName:'Sagar', Age:21, Branch:'Civil',
    Address:{ City:'Aurangabad', State:'Maharashtra' },
    Hobbies:['Cycling','Travel'] },
  { RollNo:6, StudentName:'Aisha', Age:17, Branch:'IT',
    Address:{ City:'Pune', State:'Maharashtra' },
    Hobbies:['Reading','Yoga'] },
  { RollNo:7, StudentName:'Kiran', Age:23, Branch:'Computer',
    Address:{ City:'Mumbai', State:'Maharashtra' },
    Hobbies:['Music','Cricket'] },
  { RollNo:8, StudentName:'Neha', Age:20, Branch:'Electronics',
    Address:{ City:'Nagpur', State:'Maharashtra' },
    Hobbies:['Cooking','Reading'] },
  { RollNo:9, StudentName:'Arjun', Age:16, Branch:'Civil',
    Address:{ City:'Pune', State:'Maharashtra' },
    Hobbies:['Football','Swimming'] },
  { RollNo:10, StudentName:'Meena', Age:22, Branch:'IT',
    Address:{ City:'Kolhapur', State:'Maharashtra' },
    Hobbies:['Drawing','Travel'] }
])
```

4. Display all student information

```
db.Students.find()
```

5. Students with age > 15

```
db.Students.find({ Age: { $gt: 15 } })
```

6. Students sorted by name

```
db.Students.find().sort({ StudentName: 1 })
```

7. Update branch of RollNo 3 to Computer

```
db.Students.updateOne(
  { RollNo: 3 },
  { $set: { Branch: 'Computer' } })
```

```
|)
```

8. Remove document with RollNo 1

```
|db.Students.deleteOne({ RollNo: 1 })
```

9. Students whose name starts with 'A'

```
|db.Students.find({ StudentName: /^A/ })
```

10. Total count of documents

```
|db.Students.countDocuments()
```

11. Display only first 5 documents

```
|db.Students.find().limit(5)
```

12. Display all documents except first 3

```
|db.Students.find().skip(3)
```

13. Students living in Pune

```
|db.Students.find({ 'Address.City': 'Pune' })
```

14. Display only name of all students

```
|db.Students.find({}, { _id: 0, StudentName: 1 })
```

15. Drop collection

```
|db.Students.drop()
```